# Perception by Generation: Evaluating Emergent Robotic Understanding in Video World Models

Midterm Report

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## 1 Project Overview

Video world models are increasingly proposed as simulators for robot learning, but they are usually evaluated on whether a generated clip completes the task, not on whether the clip represents the scene correctly. A model can therefore produce a convincing video while getting the underlying scene wrong. This project measures what a video world model understands about a robot scene by prompting it to draw that understanding directly (for example, as a segmentation overlay or a depth map) and scoring the result against ground truth, building on recent work that treats generation as a route to perception [1, 4]. The aim is not to outperform specialist perception models, but to find out whether the scene structure a robot simulator needs is present at all, how far it lags behind specialist models, and how it fails.

The four objectives from the proposal are unchanged:

- **O1:** define a taxonomy of scene properties to measure;
- **O2:** design a scoring protocol that turns a generated clip into a quantitative measurement;
- **O3:** apply the protocol to several world models on real robot observations;
- **O4:** build a taxonomy of the ways these readouts fail.

One addition since the proposal is an instruction-grounding step. Each instruction is parsed once into the objects it mentions, and both the prompt and the ground truth are restricted to those objects.

## 2 Progress Made

Table 1 summarises the status of the seven phases from the proposal.

The main change from the proposal is the order of work: the evaluation code was completed and stabilised before scaling up the data. As a result, adding a new task or a new world model now mostly requires configuration rather than changes to the pipeline.

## 3 Implementation

### 3.1 System design

The system has two model-agnostic pipelines:

| Phase | Status |
|---|---|
| Scoping and literature review | Complete. |
| Data curation | Complete. Full benchmark ingested; 20-item pilot subset fixed. |
| Task taxonomy and feasibility pilot | Six of seven tasks implemented; five scored end to end. Contact detection deferred. |
| Evaluation protocol | Complete, including fidelity checks and per-scenario breakdowns. |
| Model evaluation | Pilot run completed for Veo 3.1; other models pending. |
| Results and analysis | Preliminary results below; full analysis and failure taxonomy pending. |
| Writing and release | Not started. |

Table 1: Status of each project phase.

- **Generation (`wmgen`)** takes the conditioning frame and a task-specific prompt and generates a video clip. Each world model is added through a configuration file, and generated clips are cached so that no clip is generated (and paid for) twice.
- **Evaluation (`wmeval`)** takes each generated clip and its original frame and produces scores in four stages: ground-truth derivation, frame selection, answer extraction, and metric computation. Ground-truth outputs from the specialist models are cached, so metrics can be revised without re-running those models.

Before a score is accepted, three *fidelity checks* are applied to each clip: whether the scene outside the marked region is preserved, whether the requested overlay was actually drawn, and whether the output is stable rather than flickering between frames. A failed check does not change the score; instead, the sample is flagged, and the proportion of flagged samples is reported as an *invalid rate* alongside every result.

### 3.2 Data

The full RoboTrustBench dataset [2] has been ingested: 1,207 expert-validated instruction–image pairs drawn from DROID [3], covering four instruction scenarios (Normal, Constraint-Sensitive, Counterfactual, and Adversarial). For the feasibility pilot, a 20-item subset was sampled in proportion to the scenario distribution, giving 9 Normal, 6 Constraint-Sensitive, 4 Counterfactual, and 1 Adversarial item.

### 3.3 Tasks

Each task is implemented as a self-contained probe that defines the prompt (what the model is asked to draw), the extractor (how the drawing is decoded), the ground-truth model, and the metrics. Keeping the prompt and extractor in the same module ensures they use the same visual convention, so a mismatch between them is not mistaken for a model failure. Table 2 lists the seven tasks.

## 4 Scoring Method

All tasks follow the same procedure. First, the world model is prompted to draw a specific visual output, such as a colour-coded overlay, a depth map, or a set of coloured dots. Second, the ground truth is produced by running a specialist model on the original conditioning frame only, so it is

| Level | Task | Ground truth | Status |
|---|---|---|---|
| Scene | Semantic segmentation | SAM 3 [5] | Scored |
| Scene | Depth perception | Depth Anything 3 [6] | Scored |
| Interaction | Grasp region segmentation | AnyGrasp | Implemented, ground truth pending |
| Interaction | Contact detection | Gripper state | Deferred |
| Interaction | Active object detection | SAM 3 | Scored |
| Robot state | Robot keypoint detection | RoboPEPP | Scored |
| Robot state | Robot pose estimation | RoboPEPP | Scored |

Table 2: Task taxonomy and current status.

never influenced by the generated video. Third, a single frame of the generated clip (the *readout frame*) is decoded into the same format as the ground truth, such as masks, a depth map, or a set of points. Finally, the two are compared using task-specific metrics. As an upper bound, each task is also scored on the best-matching frame in the clip (the *oracle frame*). The five scored tasks are described below. Grasp region segmentation is also implemented, but its ground truth is still pending as the model to be used is not decided upon as of yet.

**Semantic segmentation.** The model is asked to paint each task-relevant object in its own solid colour and leave the rest of the image unchanged. Ground truth is a set of SAM 3 masks for the objects named in the instruction. Masks are extracted from the final frame by separating strongly coloured pixels by hue. Predicted and ground-truth masks are matched one-to-one, and the main metric is mean IoU, computed by dividing the sum of matched IoUs by the *larger* of the two object counts. This prevents a model from scoring well by painting only one object accurately and ignoring the rest. Precision, recall, and panoptic quality are also recorded.

**Depth perception.** The model is asked to replace the image with a greyscale depth map, with near surfaces bright and far surfaces dark. Ground truth comes from Depth Anything 3. Since a depth map can drift over time, the readout is the most stable frame near the end of the clip rather than the last frame. Brightness is converted back to depth, and because both maps give only relative depth, a scale and offset are fitted before comparison. The main metric is $\delta^1$, the fraction of pixels whose predicted depth is within a factor of 1.25 of the ground truth.

**Active object detection.** The model is asked to mark only the object the robot is acting on. Ground truth is the SAM 3 mask of the instruction's target object. In addition to the segmentation metrics, this task reports a *centroid hit* rate: the fraction of cases where the predicted mask covers the centre of the correct object. This directly measures whether the model identified the right object, even if its mask outline is imprecise.

**Robot keypoint detection.** The model is asked to place a coloured dot on each joint of the robot arm. Ground truth is the set of 2-D joint positions predicted by RoboPEPP. The dots are matched one-to-one to the ground-truth joints, and the main metric is PCK: the fraction of joints with a predicted dot within 5% of the image diagonal.

**Robot pose estimation.** The model is asked to mark the centre of the gripper with a red dot and a point slightly ahead of it, along the direction the gripper points, with a green dot. Ground truth is derived from two RoboPEPP keypoints at the end of the arm. The pose score combines

position and orientation error:

$$\text{pose score} = \exp\left(-\frac{\text{position error}}{\tau}\right) \times \left(1 - \frac{\text{angle error}}{90^\circ}\right),$$

where $\tau$ is a distance threshold. The score is zero if either dot is missing.

# 5 Preliminary Results

A pilot run was completed with Veo 3.1, generating 120 clips (six tasks across 20 items). Five tasks have been scored. For three items, ground truth could not be produced for the object-based tasks: two instructions refer to the object only as "it", and SAM 3 could not locate the "black hoodie" named in the third. These items are excluded from those tasks, which is why $n = 17$ in some rows of Table 3.

| **Task (metric)** | **Level** | $n$ | **Invalid** | **Readout** | **Oracle** |
|---|---|---|---|---|---|
| Semantic segmentation (mean IoU) | Scene | 17 | 35% | 0.114 | 0.193 |
| Depth perception ($\delta^1$) | Scene | 20 | 0% | 0.359 | 0.400 |
| Active object (mean IoU) | Interaction | 17 | 29% | 0.088 | 0.143 |
| Robot keypoints (PCK) | Robot state | 20 | 35% | 0.264 | 0.479 |
| Robot pose (pose score) | Robot state | 20 | 10% | 0.057 | 0.080 |

Table 3: Pilot results for Veo 3.1 on the readout frame and the oracle frame.

## 5.1 Observations

With only 20 items and one model, these results are intended to show that the pipeline works end to end rather than to support firm conclusions. Two observations are nonetheless worth noting.

**All scores are well below the specialist models.** Every main metric is far below 1, the value a perfect reproduction of the specialist model's output would receive. For the object-level tasks in particular, mean IoU is around 0.1 for both semantic segmentation (0.114) and active object detection (0.088). Without further prompting or adaptation, Veo 3.1 recovers only a small part of the scene structure these tasks measure.

**Correct answers often appear before the final frame for keypoints.** For robot keypoints, PCK rises from 0.26 on the readout frame to 0.48 on the oracle frame, the largest gap of any task. Because the oracle frame is chosen using the ground truth, it is an optimistic upper bound and part of this gap is expected from selection alone. However, the gap is clearly larger than for the other tasks, which suggests that the model often places the joint markers correctly at some point in the clip and then moves away from them. Whether this reflects genuine drift will be checked during the full evaluation, as it would be a candidate failure mode for O4.

The invalid rates in Table 3 show that 29–35% of the segmentation, active object and keypoint clips fail at least one fidelity check. A breakdown of which check is responsible has not yet been produced, so these rates are reported here without further interpretation.

## 6 Next Steps

The pilot showed that not every RoboTrustBench item is suitable for this kind of evaluation. Some instructions do not name their target object, and some objects cannot be reliably grounded by the specialist models. The plan is therefore changing from evaluating the full 1,207-item set to building a curated benchmark for robotic scene understanding in video models: a subset of RoboTrustBench in which every item has verified ground truth for every task and is scored through one fixed pipeline. Video models will then be evaluated on this benchmark using the protocol described in this report. The remaining work is as follows.

1. **Curate the benchmark subset.** Select items whose instructions name a clear target object and whose objects the specialist models can ground, while keeping enough items in each scenario for a meaningful comparison between scenarios.

2. **Verify the ground truth.** Check the specialist model outputs for each selected item, and exclude items whose ground truth is unreliable so that they are not counted as model failures.

3. **Complete the remaining tasks.** Finish grasp region segmentation once a ground-truth model is confirmed, with open alternatives to AnyGrasp (GSNet, GraspGen, Contact-GraspNet) under consideration . Decide on a ground-truth source for contact detection, since RoboTrust-Bench does not provide gripper-state labels.

4. **Fix the pipeline.** Calibrate the fidelity-check thresholds, which are currently provisional, and report how sensitive the invalid rate is to them. Then keep prompts, readout rules and thresholds fixed so that every model is scored in the same way.

5. **Evaluate multiple models.** Score open-weight generators (Wan2.2, HunyuanVideo) and other commercial models alongside Veo 3.1, and report results by task, by level and by scenario.

6. **Analyse failures and release.** Use the results to build the failure taxonomy (O4) and release the benchmark subset, ground truth and scoring code.

In summary, the pipeline and scoring protocol are in place, and one model has been evaluated on five tasks. The focus for the rest of the project is turning this pipeline into a curated benchmark with verified ground truth, completing the remaining tasks, and using it to compare video models.

## References


[1] T. Wiedemer et al., *Video models are zero-shot learners and reasoners*, arXiv:2509.20328, 2025.

[2] H. Li et al., *RoboTrustBench: Benchmarking the Trustworthiness of Video World Models for Robotic Manipulation*, arXiv:2606.01600, 2026.

[3] A. Khazatsky et al., *DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset*, arXiv:2403.12945, 2025.

[4] M. King et al., *Gen4U: Unifying Video Generation and Understanding via Diffusion*, arXiv:2607.06856, 2026.

[5] N. Carion et al., *SAM 3: Segment Anything with Concepts*, arXiv:2511.16719, 2026.

[6] H. Lin et al., *Depth Anything 3: Recovering the Visual Space from Any Views*, arXiv:2511.10647, 2025.